

HelioGuard AI

NASA Space Risk Intelligence Dashboard

A professional research proposal for a Stardance flagship project using NASA public/open data, mathematical risk scoring, and explainable AI-style mission briefings.

Research Positioning

HelioGuard AI is not framed as an official warning system. It is an educational intelligence layer that helps students interpret asteroid and space weather data with transparent scoring and readable explanations.

Project type	NASA data dashboard + AI-style reasoning + mathematical model
Target audience	Students, teen builders, science learners, and non-experts
Build stack	React, Vite, TypeScript, NASA Open APIs, rule-based briefing engine
Primary output	Live web dashboard, GitHub repo, research document, Stardance devlogs

1. Executive Summary

HelioGuard AI is a NASA-data-powered educational dashboard that transforms asteroid and space weather information into mathematical risk scores and clear mission briefings. The project combines public NASA data, transparent formulas, and a rule-based AI-style explanation engine to make complex space data easier to understand.

The product is designed as a flagship Stardance build: it is visually strong, technically meaningful, and explainable. It avoids claiming official NASA authority. Instead, it clearly states that it uses NASA public/open data and that the scores are educational approximations.

Core Thesis

Raw scientific data becomes more valuable to young builders when it is converted into structured features, normalized scores, visual risk levels, and plain-English reasoning.

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2. Problem Statement and Motivation

NASA and space weather agencies publish valuable scientific data, but raw API output is difficult for beginners to interpret. A student may see asteroid diameter, miss distance, velocity, solar flare class, or geomagnetic storm metadata without understanding what those values mean in context.

The project addresses this interpretation gap by building an educational pipeline from raw data to readable insight. NOAA SWPC explains that different types of space weather can affect technologies on Earth, including radio communication, satellite electronics, GPS/GNSS accuracy, and electric power infrastructure. This makes space weather a strong topic for a dashboard that connects science to real technology impact.

User pain point	HelioGuard AI response
Raw NASA data is difficult to read.	Convert fields into clear cards, labels, and explanations.
Risk is not obvious from raw numbers.	Create transparent 0-100 educational scores.
Space weather feels abstract.	Explain GPS, satellites, radio, and power-grid relevance.
Students need a project story.	Use a research-backed build narrative and clear methodology.

Important framing

This proposal does not claim that HelioGuard AI can predict disasters. The model is educational: it helps users understand why certain values matter, not replace NASA, NOAA, or professional warnings.

3. Research Questions and Hypothesis

Main research question: How can NASA asteroid and space weather data be transformed into a student-friendly risk intelligence dashboard using mathematical scoring and explainable AI-style reasoning?

Hypothesis: A student-friendly dashboard can make NASA space data easier to understand by combining public APIs, feature extraction, normalization, transparent scoring, visual dashboard design, and rule-based plain-English summaries.

The first version should prioritize transparency over complexity. A rule-based model is easier to verify than a black-box AI model and is safer for an educational demo. Later versions can add optional LLM summarization after the mathematical layer is stable.

4. Data Sources

HelioGuard AI should start with official and well-documented public sources. The project can launch with NASA Open APIs and then add NOAA context pages for educational explanations.

Source	Use in HelioGuard AI
NASA NeoWs	Near-Earth asteroid feed: name, estimated diameter, close approach date, relative velocity, miss distance,
NASA DONKI	Space weather events: solar flares, CMEs, geomagnetic storms, notifications, and related event metadata.
NOAA SWPC	Educational context for space weather impacts on radio, GPS/GNSS, satellites, and electric power systems.

NASA Open APIs describes NeoWs as a RESTful web service for near-Earth asteroid information. NASA API documentation also lists DONKI as a space weather database/API. NOAA SWPC impact pages provide context for how space weather affects Earth technologies.

Data integrity rule

Every dashboard card should show the source type and timestamp when possible. If data fetch fails, the app should show a clean fallback state instead of pretending the data is current.

5. System Architecture

The architecture separates data access, transformation, scoring, and explanation. This makes the project easier to test and easier to explain in a README or Stardance devlog.

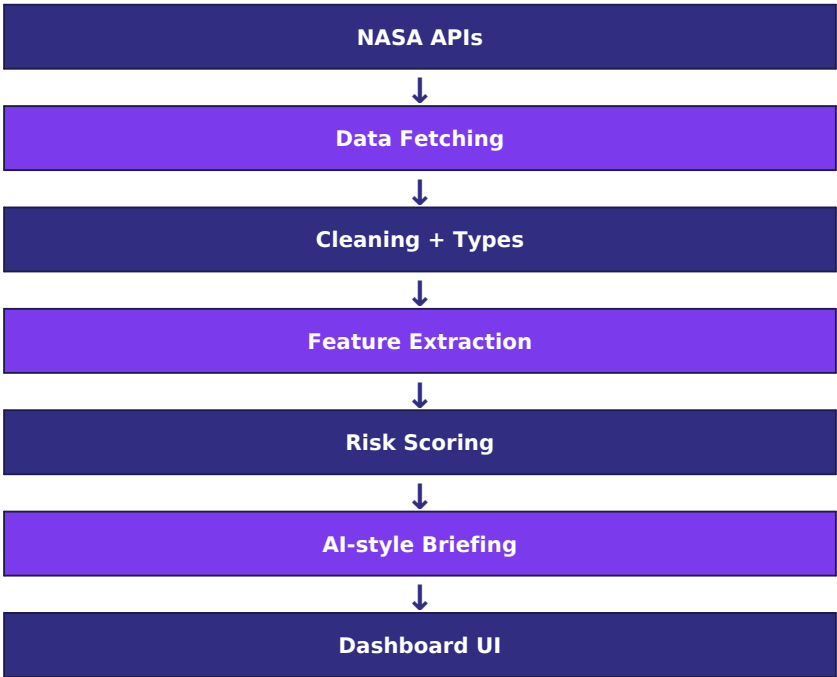


Figure 1. High-level data-to-briefing pipeline.

Layer	Responsibility
API service layer	Fetch NASA NeoWs and DONKI data using clean TypeScript service functions.
Type mapping	Convert raw JSON into typed objects that the app can trust.
Feature extraction	Extract diameter, velocity, distance, flare class, event counts, and flags.
Scoring engine	Normalize features and compute AsteroidRisk, SpaceWeatherRisk, and HelioGuardIndex.
Briefing engine	Generate plain-English mission summary from scores and top events.
Dashboard UI	Render cards, timelines, badges, sources, loading states, and disclaimers.

6. Mathematical Risk Model

The mathematical model is intentionally transparent. It uses normalized feature values and weighted combinations to produce an educational score from 0 to 100. The exact weights are starting assumptions, not official scientific hazard calculations.

6.1 Normalization

Raw values have different units: asteroid diameter is measured in meters, velocity in km/s, miss distance in kilometers or lunar distance, and hazard status is a boolean. Normalization converts these values to a comparable 0-1 range.

```
normalizeRange(x, min, max) = clamp((x - min) / (max - min), 0, 1)
```

6.2 Asteroid Risk Score

For each near-Earth object, define four features:

```
D = estimated diameter in meters V = relative velocity in km/s M = miss distance H = 1 if NASA marks the object potentially hazardous, otherwise 0
```

Distance is inverted because a smaller miss distance indicates a closer approach. The MVP formula is:

AsteroidRisk = 100 * clamp(0.30 * S_diameter + 0.20 * S_velocity + 0.35 * S_distance + 0.15 * H, 0, 1)

Weight	Reasoning
Distance - 35%	Closer approaches deserve more attention in an educational risk visualization.
Diameter - 30%	Larger objects have greater physical significance.
Velocity - 20%	Kinetic energy grows with velocity squared.
NASA hazard flag - 15%	A derived signal from the data source should influence the score.

Physical intuition

Kinetic energy follows $E = \frac{1}{2} * m * v^2$. For a roughly spherical object, mass scales with volume, and volume scales with diameter cubed. This means an educational impact-energy proxy can be described as $D^3 * V^2$.

6.3 Space Weather Risk Score

Space weather risk is event-based rather than object-based. The MVP can score the presence and severity of solar flares, CMEs, geomagnetic storms, solar energetic particles, and notification density.

F = solar flare score C = CME score G = geomagnetic storm score S = solar energetic particle score N = notification density score

SpaceWeatherRisk = 100 * clamp(0.30 * F + 0.25 * C + 0.25 * G + 0.10 * S + 0.10 * N, 0, 1)

6.4 Combined HelioGuard Index

HelioGuardIndex = 0.45 * AsteroidRisk + 0.55 * SpaceWeatherRisk

Space weather receives slightly higher weight because it is more directly connected to daily technology systems such as satellites, GPS, radio communication, and power infrastructure. Near-Earth asteroid monitoring is important, but most close approaches do not produce immediate daily technology effects.

6.5 Risk Bands

Score range	Label	Interpretation
0-24	LOW	Normal conditions. Educational monitoring only.
25-49	MODERATE	Some activity detected. Minor awareness recommended.
50-74	ELEVATED	Multiple risk signals detected. Explain possible technology relevance.
75-100	HIGH	Strong event cluster detected. Show official-source disclaimer prominently.

Disclaimer to show in the app

HelioGuard AI is an educational dashboard, not an official emergency alert system. For real alerts or operational decisions, check official NASA, NOAA, and government sources.

7. AI-Style Reasoning Engine

The MVP should use a rule-based AI-style briefing engine. This makes the project transparent, deterministic, and easy to debug. The engine converts scores and top events into a short mission briefing.

Input	Example use
HelioGuardIndex	Choose overall briefing tone and risk label.
AsteroidRisk	Mention largest/closest object and whether the approach appears safe.
SpaceWeatherRisk	Mention radio, GPS, satellite, and power-grid relevance if score is elevated.
Top events	Summarize the most important asteroid or DONKI event.
Data source status	Tell the user if NASA data is live, cached, or unavailable.

IF HelioGuardIndex < 25: explain conditions appear calm IF SpaceWeatherRisk >= 50: mention technology impact categories IF AsteroidRisk >= 50: mention closest object, size, velocity, and miss distance IF data fetch fails: show limitation and source status IF risk is high: remind users to check official sources

Example Mission Briefing

Today's HelioGuard Index is MODERATE. NASA asteroid data shows several near-Earth objects passing safely, while space weather activity remains limited. Current technology impact risk appears low, but solar activity should still be monitored because stronger events can affect GPS, radio communication, satellites, and power systems.

8. UI/UX and Product Requirements

The interface should feel like a mission-control dashboard while staying readable. The goal is not just decoration; the UI should make the reasoning visible.

Area	Requirement
Visual style	Dark space theme, blue/purple glow, readable typography, clean cards.
Hero	Project title, one-liner, data source badges, current HelioGuard Index preview.
Asteroid Watch	Cards for name, size, velocity, miss distance, hazard flag, and risk badge.
Space Weather	Timeline cards for DONKI events and technology impact notes.
Mission Briefing	Terminal-like summary with plain-English explanation.
Methodology	Short math model explanation to boost trust and technical credibility.
States	Loading, error, empty, data-source unavailable, and responsive mobile states.

Suggested pages/sections:

- Hero and current risk index
- Asteroid Watch
- Space Weather Monitor
- AI Mission Briefing
- Methodology and data sources
- Educational disclaimer

9. Implementation Roadmap

The roadmap is designed for fast Stardance progress: ship an MVP quickly, then improve it through visible v2/v3 upgrades.

Phase	Build target	Ship value
Day 1 MVP	React/Vite setup, NASA NeoWs fetch, asteroid cards, first risk score.	Assembling NASA data dashboard.
Day 2	Mission briefing engine, HelioGuard Index, error/loading states.	Clear AI-style reasoning story.
Day 3	DONKI service, space weather timeline, README, screenshots, deployment.	Ship ready Stardance submission.
v2 Upgrade	Charts, date selector, saved briefings, methodology panel.	Higher technicality and usability.
v3 Flagship	Student explanation mode, optional LLM summarizer, download/print.	Model originality and storytelling.

Cursor build strategy

Ask Cursor to implement in small steps: first the data service, then risk utilities, then components, then UI polish. Do not let it rewrite the whole app at once.

10. Evaluation Plan

The project should be evaluated both as software and as an educational research artifact.

Metric	How to evaluate
Correct data loading	NeoWs and DONKI service calls return typed objects or clean fallback states.
Score transparency	Every risk score can be traced to visible features and formulas.
Briefing quality	Briefing mentions risk level, reason, technology relevance, and disclaimer.
Usability	A non-expert can understand the dashboard in under one minute.
Stardance readiness	Live demo, screenshots, README, devlogs, and GitHub repo are complete.

11. Safety, Ethics, and Limitations

Because this project discusses risk, it must be careful with claims. The dashboard should not create panic or imply operational authority. The app should always distinguish between educational risk visualization and official hazard forecasting.

- Do not claim official NASA partnership unless officially stated by the event or source.
- Do not label scores as real emergency predictions.
- Always show source links and update timestamps when possible.
- If live data fails, state the limitation clearly.
- Keep the model weights visible so users can inspect the reasoning.

12. Stardance Positioning

One-liner

A NASA-data-powered AI dashboard that turns asteroid and space weather data into mathematical risk scores and mission briefings.

Project description

HelioGuard AI uses NASA public/open data to make asteroid and space weather information easier to understand. It extracts important data features, computes educational risk scores, and generates plain-English mission briefings so students can understand what is happening in space and why it matters for Earth technology.

First devlog draft:

Today I started HelioGuard AI, a NASA-data-powered space risk intelligence dashboard. I researched NASA Open APIs and planned the main modules: Asteroid Watch, Space Weather Monitor, mathematical risk scoring, and AI-style mission briefing. The goal is to turn raw space data into a clear dashboard that students can understand and explore.

13. References

- [1] NASA Open APIs
<https://api.nasa.gov/>
Used for NASA NeoWs and DONKI API documentation.
- [2] NASA Open Data Portal - Asteroids NeoWs API
<https://data.nasa.gov/dataset/asteroids-neows-api>
Source description for Near-Earth Object Web Service.
- [3] NASA - Hack Club Stardance Challenge
<https://www.nasa.gov/stem-content/hack-club-stardance-challenge/>
Context for Stardance challenge and use of NASA mission data.
- [4] NOAA SWPC - Space Weather Impacts
<https://www.swpc.noaa.gov/impacts>
Educational context for effects on radio, satellites, and other technologies.
- [5] NOAA SWPC - Space Weather and GPS Systems
<https://www.swpc.noaa.gov/impacts/space-weather-and-gps-systems>
Context for GPS/GNSS relevance.
- [6] NOAA SWPC - Electric Power Transmission
<https://www.swpc.noaa.gov/impacts/electric-power-transmission>
Context for power grid relevance.
- [7] NOAA SWPC - NOAA Space Weather Scales
<https://www.swpc.noaa.gov/noaa-scales-explanation>
Context for public communication of space weather severity.

Prepared for: BeBe / Nero_404. Suggested repository: github.com/BeBecpp/helioguard-ai. Suggested deployment: Vercel or GitHub Pages.

Document status: Research proposal and build brief, ready to be adapted into README.md and docs/RESEARCH.md.